

[TODO: Model name]: Bi-Axial Attention and Register-Carried Cascades for In-Context 3D Medical Segmentation

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Abstract

*In-context learning (ICL) allows a segmentation model to adapt to a new anatomical structure at inference time from a handful of (image, mask) context pairs, without any gradient update. Extending ICL segmentation to full 3D volumes is attractive for clinical CT/MRI workflows but exposes a resolution–compute trade-off that is more severe than in 2D: dense cross-attention between a target volume and its contexts scales cubically with side length, forcing prior work to either downsample the volume, fall back to per-slice processing, or tile the volume with independent, context-blind sliding windows. We propose a 3D in-context segmentation model built around two components. First, a bi-axial image–label attention mechanism lets target and context tokens exchange information jointly along the spatial-patch axis and the image/label axis, in contrast to early feature concatenation (Medverse [?]) or one-shot pixel-shuffle fusion (Iris [?]). Second, a coarse-to-fine cascade initializes each finer level’s label tokens from the previous level’s up-sampled prediction and restricts its sliding window to the previously predicted foreground region, while a small set of learned register tokens carries hidden state across levels so the whole cascade can be trained end-to-end with gradients flowing between resolutions. We evaluate on TotalSegmentator [?] CT at a fixed 3 mm spacing, reporting Dice, normalized surface distance, and compute (wall-clock, FLOPs, peak VRAM) against Medverse and Iris baselines, and ablate the query prior, the region-restricted fine level, and cross-level register training. **[TODO: fill in headline numbers once experiments are final: e.g., X% Dice at Y% of the compute of the strongest baseline, and standout categories such as small/thin structures.]** We further test generalization to **[TODO: other CT cohorts, MRI, and far-OOD tasks]**, and release code and checkpoints.*

1. Introduction

In-context learning (ICL) recasts medical image segmentation as a few-shot matching problem: given a target scan and a small set of (image, mask) context pairs illustrating the structure of interest, the model predicts a mask for the target without any task-specific fine-tuning. This removes the need to retrain or fine-tune a model for every new anatomical class, which is especially valuable for structures that are rare, newly defined, or only annotated by a handful of clinical sites.

Most ICL segmentation models were designed and validated in 2D [? ? ?]. Moving to native 3D volumes is attractive – CT and MRI are inherently volumetric, and slice-wise processing discards through-plane context that is often necessary to resolve thin or elongated structures – but it sharpens an already difficult resolution–compute trade-off. Dense cross-attention between a target volume and K context volumes scales with the cube of the side length, so a design that is affordable at 64^3 becomes impractical at 256^3 . Existing 3D-capable approaches each make a different concession: Medverse [?] processes the whole volume through a sliding window at every resolution step of its autoregressive cascade, re-visiting the entire volume even once coarse levels have already localized the target; Iris [?] fuses context information into the target through a single-shot pixel-shuffle/unshuffle operation with no iterative refinement across scales. Our own concurrent 2D work on hierarchical patch selection [?] showed that supervising which regions to process, rather than relying on attention alone to discover them, yields a favorable accuracy/compute trade-off in 2D; here we ask whether the same principle extends to full 3D volumes, and how the image–label interaction itself should be designed once both axes (space and image/label identity) matter jointly.

We propose a 3D in-context segmentation model with two components that address these gaps directly.

Bi-axial image–label attention. Instead of concatenat-

ing image and mask channels before a shared encoder (Medverse) or fusing them once via pixel shuffle-unshuffle (Iris), we let target and context tokens attend jointly along two axes: a spatial-patch axis (as in standard within-volume/cross-context attention) and an image/label axis that lets label information at a given location refine, and be refined by, the corresponding image features. This keeps the image and label streams distinct for longer while still allowing early, repeated exchange between them.

Register-carried coarse-to-fine cascade. The model processes a volume through a small number of resolution levels. Two design choices connect the levels: (1) a *query image prior* – the previous level’s upsampled prediction initializes the target’s label tokens at the next level, rather than entering the network as a separate perturbed-GT branch (Medverse) or not at all (Iris); and (2) *region-restricted fine processing* – once a coarse level localizes the structure, the sliding window at finer levels is restricted to the predicted zone instead of re-scanning the full volume. To let the cascade be trained as a single model rather than a chain of independently-supervised stages, we introduce a small number of learned *register tokens* that carry hidden state from one level’s processing into the next, giving the optimizer a gradient path that crosses resolution levels.

Contributions. (1) A bi-axial image–label attention mechanism for in-context 3D segmentation that attends jointly over space and image/label identity, contrasted directly against early-concatenation and one-shot fusion baselines. (2) A coarse-to-fine cascade in which the query prior and the region restriction at fine levels are ablated independently, quantifying their individual contribution to the accuracy/compute trade-off. (3) Register-token cross-level training that lets gradients flow across the cascade, evaluated against per-level-only supervision. (4) An evaluation on TotalSegmentator [?] at fixed spacing against Medverse and Iris on accuracy (Dice, NSD) and compute (time, FLOPs, VRAM), together with generalization experiments on [TODO: other CT datasets, other modalities, and far-OOD tasks].

2. Related Work

In-context medical image segmentation. UniverSeg [?] popularized ICL segmentation with a CrossBlock architecture that densely cross-attends target and context features at full resolution. Iris [?] instead encodes each context pair into a compact task embedding and fuses it into the target features through pixel shuffle-unshuffle operations, avoiding dense cross-attention but performing the image–label fusion in a single shot with no coarse-to-fine refinement. Both were designed for 2D slices or small, fixed-size 3D patches; naive extension to full 3D volumes reintroduces the cubic scaling that motivated patch- and cascade-based

alternatives.

Coarse-to-fine and autoregressive cascades. Medverse [?] performs next-scale autoregressive ICL: at each step, the target image and the previous step’s prediction form an “autoregressive context” that is processed, together with the semantic (image, mask) context, by weight-shared U-Net branches, following the next-scale prediction idea of VAR [?]. Crucially, every autoregressive step still performs a full sliding-window pass over the entire volume; only the *input* to that pass changes with scale, not its spatial extent. Deep Neural Patchworks [?] similarly stacks levels of increasing resolution but targets single-model supervised segmentation rather than in-context, few-shot prediction. In contrast, our cascade restricts the sliding window itself to the region already localized by the previous level, so compute at fine levels scales with the size of the structure rather than the size of the volume, and connects levels with register tokens rather than by feeding back only the rendered prediction.

Image–label fusion mechanisms. How a model combines image and label information from the context is a central design axis. Concatenation-based fusion (Medverse, and earlier 2D CrossBlock-style models) merges the two streams early, before most of the network’s capacity is applied. Iris fuses them once via a pixel-shuffle/unshuffle operation between context and target. We instead treat image/label identity as a first-class attention axis, alongside the spatial-patch axis, allowing the two streams to exchange information repeatedly across attention layers rather than once at the input or via a single fusion step. [TODO: position against additional fusion designs if relevant, e.g. gated cross-attention or FiLM-style conditioning.]

Efficient 3D volumetric backbones. Our encoder builds on the nnU-Net residual encoder [?] and 3D axial rotary position embeddings for within-volume and cross-context self-attention, following the general trend of combining convolutional feature extraction with attention-based context aggregation in volumetric segmentation [? ?]. [TODO: expand with any additional 3D-specific efficiency baselines we compare against.]

3. Method

Input. A target volume $I^t \in \mathbb{R}^{D \times H \times W}$ and a context set $\{(I_k^c, L_k^c)\}_{k=1}^K$ of image/mask pairs for the same anatomical class as the (unknown) target mask.

Output. A predicted target mask $\hat{L}^t$, produced by a coarse-to-fine cascade over M resolution levels $\{r_1, \dots, r_M\}$. ?? contrasts our architecture with Medverse’s.

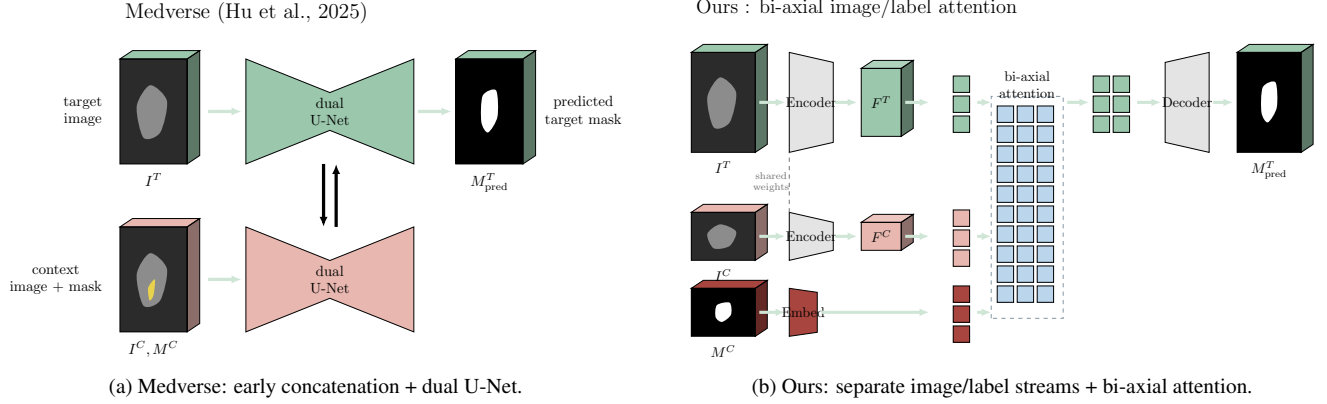


Figure 1. Architecture overview, contrasted against Medverse’s early-concatenation fusion. **[TODO: draft diagram; finalize labels/scale once the architecture is settled.]**

3.1. Backbone

Each volume (target or context) is tokenized by a shared encoder – either the nnU-Net residual-encoder stages (ResEncInContext3D) [?] or a 3D patch embedding (ViTInContext3D) – into a grid of spatial tokens with 3D axial rotary position embeddings (RoPE), following the architecture already implemented in our codebase. At the bottleneck, a two-stage attention block first lets tokens attend within their own volume (target and each context volume independently, sharing weights), then lets the target cross-attend, read-only, into all K context bottlenecks. **[TODO: restate/adjust to match final architecture diagram.]**

3.2. Bi-Axial Image–Label Attention

Rather than folding the label into the image as a second input channel before encoding (concatenation, as in Medverse) or fusing image and label features once via a pixel shuffle-unshuffle step (Iris), we keep image tokens $\mathbf{z}^{\text{img}}$ and label tokens $\mathbf{z}^{\text{lbl}}$ as separate streams and attend over two axes:

- **Patch axis:** standard spatial self-/cross-attention among tokens at the same image/label identity, as described in ??.
- **Image/label axis:** at each spatial location, the image and label tokens attend to each other, letting label evidence sharpen the corresponding image features and vice versa.

?? illustrates both axes within a single layer.

The target’s label stream starts from a query prior (??) rather than a zero/blank mask, so this axis is informative for the target as well as for the contexts. **[TODO: formalize with an equation once the exact factorization (interleaved vs. parallel bi-axial layers) is finalized; specify how many layers alternate between the two axes.]**

3.3. Coarse-to-Fine Cascade

Our model processes the target through M increasingly fine resolution levels $\{r_1 < \dots < r_M\}$ (e.g. a coarse full-volume pass followed by finer crops). Two mechanisms connect consecutive levels.

Query image prior. At level $\ell > 1$, the target’s label tokens are initialized from the upsampled prediction of level $\ell - 1$,

$$\mathbf{z}_{\ell,0}^{\text{lbl}} = \text{Embed}(\text{Upsample}(\hat{L}_{\ell-1}^t)), \quad (1)$$

instead of entering the network through a separate autoregressive branch (Medverse) or not being used at all (Iris). This makes the previous level’s belief part of the same token stream that attends bi-axially with the image, rather than a side channel fused later. ?? shows this flow across two levels.

Region-restricted fine processing. Given level $\ell - 1$ ’s prediction, we take its predicted foreground region (thresholded and padded by a margin) and restrict the sliding window at level ℓ to that region, rather than re-scanning the full volume as in Medverse’s autoregressive steps. This lets compute at fine levels scale with the size of the structure of interest rather than the size of the input volume, at the cost of depending on the coarse level’s localization being approximately correct. **[TODO: specify margin/threshold values and failure-mode handling (e.g. what happens when the coarse level misses the structure entirely).]**

3.4. Cross-Level Training via Register Tokens

To connect the cascade with a gradient path that crosses resolution levels – rather than levels being independently supervised and only chained via the rendered prediction in ?? – we add a small set of learned register tokens $\mathbf{r}_\ell \in \mathbb{R}^{N_r \times d}$ at each level. Registers participate in the bi-axial attention at level ℓ alongside image and label tokens, and their updated state is carried forward, via a learned projection and

Bi-Axial Attention (one layer)

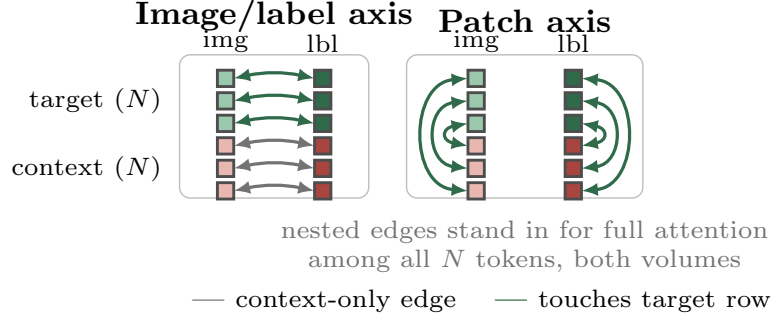
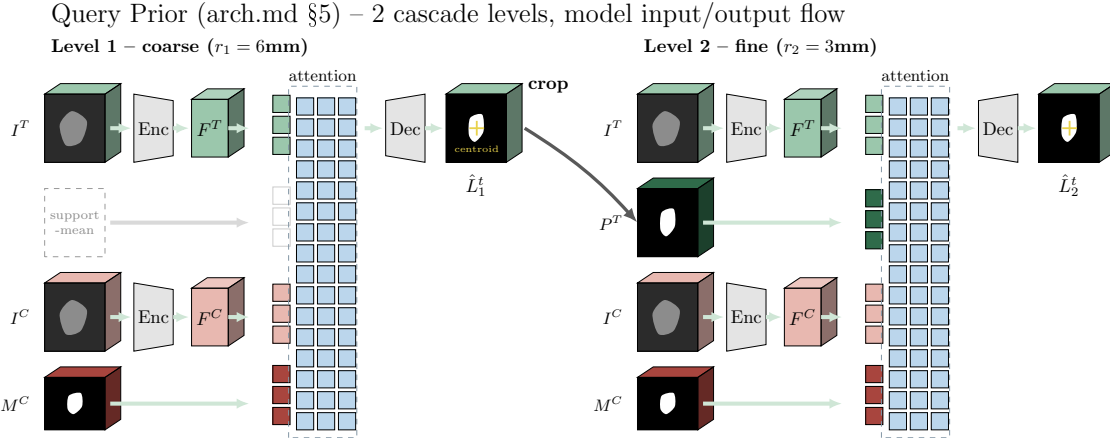


Figure 2. Bi-axial attention within one transformer layer, shown for a single context volume ($K=1$ here for clarity; in general K context volumes are each processed this way) and 3 of its N spatial tokens per stream. Left: the *image/label axis* – image and label tokens attend to each other within the same cell. Right: the *patch axis* – tokens attend across cells and, for the target, across contexts (edges stand in for full attention among all N tokens of both volumes). Gray edges are context-only; green edges touch the target row, which is architecturally distinct – it cross-attends read-only into context, and its label-stream input is the predicted query prior (??) rather than ground truth. **[TODO: draft diagram; finalize once the layer factorization above is settled.]**



level 1: target's 2nd lane = *support-mean* (no real input) → level 2: target's 2nd lane = P^T , a REAL cube (crop of $\hat{L}_1^t$) – same lane the context mask always occupied

Figure 3. Query image prior across two cascade levels, with the full model input/output flow. At level 1 the target has no real prior yet (dashed support-mean placeholder); at level 2 the target's second lane is a real crop of level 1's own prediction $\hat{L}_1^t$ – the same lane the context mask always occupies. **[TODO: draft diagram; finalize once the number of levels M is settled.]**

a level-type embedding, into level $\ell + 1$'s register initialization:

$$\mathbf{r}_{\ell+1,0} = W_r \mathbf{r}_{\ell,\text{final}} + \mathbf{e}_{\ell+1}. \quad (2)$$

This allows the model to propagate information that is useful for finer-level processing but not fully captured by the rendered segmentation mask alone (e.g. uncertainty or contextual cues about neighboring structures). ?? ablates this mechanism against training levels independently with the

query prior as the only cross-level connection.

3.5. Training Objective

Each level is supervised at its own resolution with a combined binary cross-entropy and soft Dice loss,

$$\mathcal{L}_\ell = \mathcal{L}_{\text{BCE}}^\ell + \mathcal{L}_{\text{Dice}}^\ell, \quad (3)$$

and the total loss sums over levels, **[TODO: state whether uniform or level-weighted, matching the final training]**

config]:

$$\mathcal{L} = \sum_{\ell=1}^M \lambda_{\ell} \mathcal{L}_{\ell}. \quad (4)$$

3.6. Synthetic Supervision

A fraction p_{synth} of training tasks are drawn from a GMM-based synthetic generator instead of real (image, mask) pairs: a real-anatomy supervoxel cohort is repainted with per-class Gaussian intensities, giving a label-perfect (image, mask) pair with a novel appearance. Paint noise is i.i.d. per voxel by default; an optional multi-octave correlated variant (coarse-to-fine random fields, summed) better matches the spatial autocorrelation of real tissue (measured ≈ 0.6 in real CT/MRI vs. ≈ 0.0 for i.i.d. paint at native resolution) without changing the per-class intensity variance. **[TODO: figure of a synth_gmm training pair (i.i.d. vs. multi-octave texture noise) next to a real (image, mask) pair; equation for the paint model once finalized.]**

4. Experiments

4.1. Setup

Dataset. We use TotalSegmentator [?] CT, **[TODO: state the train/held-out anatomical class split, following the same held-in/held-out protocol used in our 2D work to test in-context generalization to unseen classes, not just unseen subjects].** Experiments below are organized into two settings: (1) fixed 3 mm-spacing crops (??), isolating architectural differences in image–label interaction from the coarse-to-fine cascade; and (2) full-resolution coarse-to-fine evaluation (??), exercising the cascade end to end.

Baselines. Medverse [?] and Iris [?], **[TODO: state whether baselines are retrained on the same data/split or evaluated with released weights, and the number of context pairs K used at inference; blocker: no Iris baseline is implemented/run yet, so all current numbers are vs. Medverse only].**

Metrics. Segmentation accuracy is reported as Dice and normalized surface distance (NSD); compute is reported as wall-clock inference time, FLOPs, and peak VRAM.

4.2. Fixed-Spacing Comparison (Setting 1: 3 mm crops)

We first compare all methods at a fixed 3 mm isotropic spacing on TotalSegmentator CT crops, isolating architectural differences in image–label interaction (??) from the coarse-to-fine cascade evaluated separately under full-resolution processing in ??. **[TODO: this replaces the earlier use_crop=false whole-body protocol (notebook 20), which is stale under the current experiment organization.]**

[TODO: Discuss overall numbers, then break down by anatomical category (organs, bones, vessels, muscles)

Table 1. Dice, NSD, and compute at fixed 3 mm spacing on TotalSegmentator CT held-out classes. **[TODO: fill in.]**

Method	Dice	NSD	Time (s)	FLOPs (G)	VRAM (GB)
Iris	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Medverse	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Ours	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

Table 2. Effect of each cascade component on Dice and compute. **[TODO: fill in; one row per ablation, cumulative or leave-one-out.]**

Configuration	Dice	Time (s)	FLOPs
Full model	[TODO:]	[TODO:]	[TODO:]
– query prior	[TODO:]	[TODO:]	[TODO:]
– region restriction	[TODO:]	[TODO:]	[TODO:]
– register carry	[TODO:]	[TODO:]	[TODO:]

to identify where bi-axial attention helps most – we expect an advantage on small/thin structures, similar to the pattern observed for patch-based selection in our 2D work.]

4.3. Coarse-to-Fine Ablations (Setting 2: full resolution)

Unlike ??, this setting runs the full cascade at native/full resolution end to end, from a coarse full-volume level down to the finest region-restricted level, so that the effect of each cascade mechanism on the accuracy/compute trade-off can be measured where it actually matters. We ablate the three cascade mechanisms of ???? independently: (i) the query image prior (??) vs. a blank label-token initialization at every level; (ii) region-restricted fine processing (sliding window over the previous level’s predicted zone) vs. a full-volume sliding window at every level; and (iii) register-carried cross-level training (??) vs. independently-supervised levels connected only through the rendered query prior.

[TODO: Report the accuracy/compute Pareto trade-off as sliding-window restriction is relaxed (full volume → tight crop around the previous prediction), analogous to the resolution/FLOPs Pareto plot in our 2D paper [?].]

4.4. Synthetic Data Ablation

To isolate the effect of the GMM-based synthetic supervision described in ??, we compare three otherwise-identical training runs (single-level, non-cascade regime) that share a starting checkpoint and recipe and vary only the synthetic-data knob: real-only ($p_{\text{synth}} = 0$), synthetic with i.i.d. paint noise ($p_{\text{synth}} = 0.3$), and synthetic with the multi-octave texture noise of ??. **[TODO: report TotalSegmentator val-**

idation Dice (seen/unseen class macro split) at a shared training cutoff for all three arms, plus qualitative prediction figures per arm.]

4.5. Generalization

To test how well our model’s accuracy/compute trade-off transfers beyond the training distribution, we evaluate on three settings of increasing distribution shift.

Other CT datasets. [TODO: list held-out CT cohorts and results.]

Other modalities. [TODO: MRI (e.g. TotalSegmentator MRI [?]) evaluated with models trained only on CT.]

Far out-of-distribution tasks. [TODO: non-anatomical or pathology-driven segmentation tasks probing whether the region-restricted cascade still localizes correctly when the coarse level’s prior is unreliable.]

5. Conclusion

We presented an in-context 3D medical segmentation model that (1) attends jointly over space and image/label identity through a bi-axial attention mechanism, and (2) trains a coarse-to-fine cascade end-to-end by carrying hidden state across resolution levels with register tokens, while restricting fine-level computation to the region localized by coarser levels. [TODO: summarize headline results once finalized: accuracy vs. Medverse/Iris at fixed spacing, the individual contribution of each cascade component, and generalization findings.] Limitations and future work: [TODO: discuss dependence of region-restricted processing on coarse-level localization accuracy, scaling to more than $\{M\}$ levels, and extending register-carried training to other volumetric ICL backbones.]

Acknowledgments

[TODO: funding statement, matching our prior paper’s DFG acknowledgment if applicable.]

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